MIL-T-14363A(Ord)
19 January 1962
SUPERSEDING
MIL-T-14363(Ord)
17 August 1956

MILITARY SPECIFICATION

TURRET AND GUN CONTROL AND GUN FIRE CONTROL SYSTEM;
TANK, 90MM, GUN, M48A2, M48A2C

1. SCOPE

1.1 Scope. - The control system consists of a turret-and-gun hydraulic control system, and a direct sighting and indirect gun-laying system. The direct sighting system consists of a range finder, ballistic drive, computer, telescope, and periscope for the main gun and coaxial machine gun, and a periscope for the cupola machine gun. The indirect gun-laying system consists of an elevation quadrant and an azimuth indicator. The hydraulic control system consists of a control power pack, commander's control assembly, elevating mechanism, automatic superelevation actuator, deck clearance valve assembly, traverse gear box assembly, accumulator, pressure switch, selector switch box, pressure gage and turret lock. The system and installation hardware components consist of tubing, adapters, fittings, brackets, and related items required to complete and install the parts and assemblies.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-P-514	-	Plates; Designation, Transportation-Data, and Blank for Tank and Automotive Materiel.
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, Preservative.
MIL-L-10295	-	Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-S-10379	-	Suppression, Radio Interference, General Requirements for Vehicles (and Vehicular Sub-Assemblies).
MIL-T-45052	-	Tank, Combat, Full-Tracked; 90MM Gun, M48A2; Preparation for Storage and Shipment of.
MIL-T-45148	-	Tank, Combat, Full-Tracked; 90 MM Gun, M48A2.

STANDARDS
Federal

Federal STD NO. 595 - Colors.

FSC-1230

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Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking U. S. Military Property.
MIL-STD-193 - Painting Procedures: Tracked Vehicles and Tactical Wheeled Vehicles.
MIL-STD-417 - Rubber, Composition, Vulcanized General Purpose, Solid (Symbol and Tests).

DRAWINGS
Ordnance Corps
K7970854 - Azimuth Indicator.
K8744370 - Installation, Fire Control Instrument and Drive.

ENGINEERING PARTS LIST
8750104 - Tank 90mm Gun M48A2.
and 8750105 -

SQAP INDEX
8750104 - Supplemental Quality Assurance Provisions.
and 8750105

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications . - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

NATIONAL BUREAU OF STANDARD PUBLICATIONS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Preproduction model. - When specified (see 6.2), a preproduction model(s) of the complete turret and gun control and gun fire control system to be furnished under this specification shall be produced prior to the manufacture or fabrication of turrets in quantity. The turret and gun control and gun fire control system when complete shall be submitted to the Government for inspection tests to ascertain whether it has been constructed in accordance with drawings, specifications, and performance requirements specified herein. Turret and gun control and gun fire control system submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling as contemplated under the supply contract.

3.2 First production turret. - The first complete production turret(s) off the production line shall be submitted to the Government for examination and inspection tests to ascertain whether the turret(s) meet specifications and performance requir e-ments.

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3.3 Materials. - In addition to drawing requirements, all rubber items which are under tension or flexed in any manner shall be ozone-proofed to meet "C" test requirements of Standard MIL-STD-417.

3.3.1 Qualified products. - The assembly manufacturers shall be responsible for ascertaining that all assemblies and component parts entering into the construction of this assembly, which require product qualification in accordance with a specification are listed on respective qualified products lists (QPL) or have been approved for inclusion on such list. Manufacturer's inspection record shall specifically list such components, name of supplier(s) and number and date of applicable QPL.

3.4 Construction. - Turret, gun, and fire control system components shall be machined and assembled into complete systems in accordance with drawings listed or referred to in EPL 8750104 and EPL 8750105 and the detail specifications. All parts, assemblies, and subassemblies shall be identified in accordance with Standard MIL-STD-130.

3.4.1 Screw threads. - Unless otherwise specified, all internal and external threads, shall be in accordance with Handbook H28.

3.4.2 Hydraulic fluids. - For all ambient temperatures, the hydraulic fluid used shall conform to Specification MIL-H-6083, except that fluid used in the traverse gear box assembly shall conform to Specification MIL-L-10295.

3.4.3 Waterproofness. - All fire control systems components shall be waterproofed.

3.4.4 Durability. - Systems shall withstand a minimum of 2000 miles of cross-country operation without loss of operating efficiency and accuracy.

3.5 Performance. - With an electrical supply of 26 volts d.c. minimum, and vehicle sitting level within one degree, the turret and gun control system and gun fire control system, shall meet all requirements specified herein. Systems shall operate from full depression to full elevation and throughout 360 degrees traverse. The temperature, of metal components when operating under any environmental condition shall not exceed the range of minus 25 to plus 125°F. The maximum unbalance of the main armament shall be not more than 5,000 inch pounds. The systems shall be capable of operating at 18 to 30 volts d.c. vehicle supplied.

3.5.1 Turret and gun hydraulic control system.

3.5.1.1 Gunners power control. - Handle of gunner's control shall permit power operation of the control system by gunner, using either right or left hand or using both hands simultaneously. Handle shall contain a trigger for each hand for firing the main gun or the coaxial machine gun selectively as preselected by a separate switch box. Handle shall contain a switch for each hand for operating the magnetic brake. Operation of the handle shall not be restricted with use of arctic hand gear. Control-handle deadspot including backlash, the distance from the marked center position to the point at which turret or gun begins to move, shall not be more than 7 degrees along each traverse and elevation axis. Deadspot each side of neutral shall be equal within 1 degree along each individual axis. The handle shall be self centering and shall return to center position from any operating position.

3.5.1.2 Commander's power control . - Commander's control shall provide power direction and firing of the system equal to the gunner's control and shall take instantaneous control of the system, overriding the gunner's control, when the override

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switch is depressed. Commander's control handle shall be operable by use of one hand. With the gunner's control in neutral position, release of commanders control from any position shall cause gunner's control to regain instantaneous control over system. Movement of handle, deadspot, and centering shall be as specified in 3.5.1.1 for the gunner's power control. Operation of handle shall not be restricted by use of arctic hand gear.

3.5.1.3 Deck clearance valve. - A hydraulic valve shall be provided to automatically operate the gun at a minimum clearance angle over the rear deck of the tank, to prevent the gun from striking the sides of the tank or the super structure of the rear deck.

3.5.1.4 Superelevation actuator. - When ranging, the gun shall remain on target to within plus or minus 2 mils.

3.5.1.5 Gun sighting system. - The sighting system components shall conform to Drawings K7970854 and K8744370. The sighting system shall be synchronized with the main gun and the coaxial machine gun.

3.5.1.6 Manual traverse effort. - The average force applied tangentially to the crank to maintain turret traverse movement in either direction when tank is on level ground shall not exceed 17 pounds. No individual reading shall exceed 20 pounds. At uniform hand crank speeds below 30 r.p.m., chatter-free manual traverse shall be obtained.

3.5.1.7 Manual elevation effort. - With the gun balanced, the average force applied tangential to the hand crank to maintain gun elevation movement in either direction shall not exceed 11 pounds. No individual reading shall exceed 14 pounds.

3.5.1.8 Manual traverse response rate. - One revolution of the traverse hand crank shall rotate the turret 10 to 20 mils in either direction.

3.5.1.9 Manual elevation response rate. - One revolution of the elevation hand crank shall move the gun not less than 10.6 mils in either direction.

3.5.1.10 System backlash. - The control backlash with power off shall not exceed 1 mil movement of the turret in traverse between extreme right and left positions, nor exceed 1 mil movement of the gun between extreme depression and elevation positions.

3.5.1.11 Control transfer. - The manual traverse and elevation controls shall be usable simultaneously with power controls, in any combination. Transfer of control from manual to power and from power to manual shall be instantaneous. With controls in centered position, there shall be zero turret creep and zero gun drift when power is switched on or off. A delay of 15 seconds is permissible in initial transfer from manual to power when equipment has been inoperative for 18 hours or more.

3.5.1.12 Maximum controlled traverse velocity. - Gun shall be controllable in traverse at all speeds. With vehicle on level ground, the maximum controlled velocity at 26 volts input to hydraulic equipment shall be not less than one turret revolution per 15 seconds (4 r.p.m.) in either direction.

3.5.1.13 Traverse stability. - With tank and power control system turned on or off the turret manual traversing system shall prevent the gun and turret from drifting in either direction, from any target position more than 1 mil, exclusive of hull excursion, for 12 hours without recharging the hydraulic system.

3.5.1.14 Elevation stability. - With power control system turned on or off, the manual gun elevating system shall prevent the gun from drifting in any direction from any target position more than 1 mil, exclusive of hull excursion, for 12 hours without recharging the hydraulic system.

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3.5.1.15 System protection. - There shall be no damage to the system or any component as a result of driving the gun against elevation or depression stops by power or manual control at full speed. Power and manual control shall be regained instantaneously from stopping action.

3.5.1.16 Manual elevation supercharge. - The manual elevation hydraulic system shall maintain, for a minimum of 48 hours, a charge sufficient to elevate or depress a balanced main gun smoothly without repumping.

3.5.1.17 Range of gun in elevation. - The gun shall operate in power and manual to a minimum elevation of 19 degrees from the horizontal; and to a minimum depression of 9 degrees from the horizontal except over the vehicle rear deck.

3.5.1.18 Gun elevation speeds. - Gun shall be controllable in power elevation and depression at all speeds between 0.5 mil per second and 4 degrees per second.

3.5.1.19 Turret friction. - With the turret assembly ring gear bolted to the hull, and the turret complete with gun, gun mount, slip ring, and azimuth indicator installed, but without hand and hydraulic traversing mechanisms installed, torque required to maintain rotation of the turret at any speed with the tank on level terrain shall not exceed 200 foot pounds. Variation between maximum and minimum torque through 360 degrees rotation shall not exceed 50 foot pounds.

3.5.1.20 Operation on slope. - Gun shall be controllable in traverse, elevation, and depression by power or manual operation as specified in 3.5.1.13, 3.5.1.14, 3.5.1.22 and 3.5.1.23 when hull is pitched or canted on slopes up to 15 degrees. There shall be no chatter, except in manual traverse when turret is unbalanced.

3.5.1.21 Travel. - With the gun external travel lock and the turret lock disengaged, the turret and gun control system shall act as travel locks for the gun and turret when the vehicle is in motion. With the turret control power switch off, the gun slippage shall not exceed 5 mils in elevation or depression and the turret slippage shall not exceed 5 mils in traverse when the vehicle is operated as specified in Specification MIL-T-45148. With the hand traverse crank in or out of detent there shall be no movement of the hand traverse crank or the hand elevation crank by either turret induced reaction when the vehicle is in motion or by power operation of the turret and gun control system.

3.5.1.22 Time required to lay gun on stationary target. - The average time for a total of 4, 8, or 12 trials shall not exceed that tabulated in table I. These requirements shall be met by power and manual operation. The aiming point shall be 0.25 mil square.

Table I. - Gun-laying on stationary target

LAYOFF - mils		TIME - seconds	
AZIMUTH	ELEVATION AND DEPRESSION	POWER	MANUAL
0	± 10	2	2
25	--	3	3
25	± 10	5	5
100	--	4	4.5
400	--	5	10.5
800	--	6	--
1600	--	8	--
3200	--	13	--

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3.5.1.23 Tracking velocity. - With vehicle on level ground and a minimum voltage of 26 volts applied to the system, tracking speeds shall range from not more than 0.5 mil per second to not less than 65.2 mils per second. Transition from minimum to maximum velocity shall be smooth and gradual and the system shall not become unstable at any time. At velocities above 65.2 mils per second, turret velocities shall increase with increased control handle displacement until maximum handle displacement is reached. Turret velocity at this point shall not be less than 4 r.p.m.

3.5.1.24 Tracking accuracy. - With the tank in a stationary position, the gunner's and commander's power controls shall meet minimum requirements for target tracking as specified in table II. Target shall be square in specified sizes and shall be moved over a terrain of average flatness on a straight crossing course. When referenced range distance is not available, it is permissible to substitute another distance, preferably greater, providing that target speeds, target dimensions, and course distance are changed proportionately.

Table II. - Tracking conditions

% OF TIME ON TARGET	TRAVERSE RATE mils/sec	TARGET SIZE mils	*TARGET RANGE yard or meters	TARGET VELOCITY m.p.h.	LENGTH OF COURSE yards
90	0.5	0.25	2000	2.0	50
100	1.0	1.25	2000	4.0	100
100	3.3	1.25	2000	13.5	200
100	9.8	1.25	2000	40.0	200
100	19.6	1.25	300	12.0	200
100	32.6	1.25	300	20.0	200
100	65.2	5.00	300	40.0	200

*NOTE: Which linear system to be used is the same as that in the tank fire control system.

3.5.1.25 Time required to lay gun on moving target with established lead angles. - The time required to lay the gun on target from the established lead angle shall not exceed 3.5 seconds in any instance as specified in table III. The lead angle shall be established behind the target.

Table III. - Gun-laying on moving target

TRAVERSE RATE mils/sec	ESTABLISHED LEAD mils	TARGET SIZE mils	*TARGET RANGE yards or meters	TARGET VELOCITY m.p.h.	LENGTH OF COURSE yards
0.5	1.0	0.25	2000	2.0	50
1.0	2.0	1.25	2000	4.0	100
3.3	7.5	1.25	2000	13.5	200
9.8	20.0	1.25	2000	40.0	200
19.6	7.5	1.25	300	12.0	200
32.6	10.0	1.25	300	20.0	200
65.2	20.0	5.00	300	40.0	200

*NOTE: Which ever linear system is employed the figures indicated should be used.

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3.5.2 Gun fire control system.

3.5.2.1 Backlash. - The elevation backlash in the system shall be not more than 0.3 mil for any gun elevation throughout the entire range of elevation and depression.

3.5.2.2 Line of sight. - The lines of sight of the range finder, gunner's periscope, and telescope shall move in synchronization with the centerline of the gun tube within plus or minus 0.5 mil right or left of a plumb line and plus or minus 0.3 mil above or below all positions from 0 to 15 degrees elevation.

3.5.2.3 Mounting. - The gunner's periscope and range finder shall function freely, without deflection within the full range from maximum gun elevation to maximum depression.

3.5.2.4 Boresight knob travel. - The boresight knobs on range finder, gunner's periscope, and telescope mount shall be adjustable to a minimum of 5 mils down, 2 mils up, and 4 mils left and right, except the coincidence range finder which shall be 3.65 mils left and 4.35 mils right as measured from the boresight position specified.

3.5.2.5 Leveling devices. - The levels on the range finder and ballistics drive shall be level within 15 minutes of arc when the gun is level and the ballistic drive superelevation input shaft is at the qualified position.

3.5.2.6 Coincidence range finder calibration. - Each coincidence range finder will be checked immediately prior to installation in the turret and after installation of the turret in the vehicle.

3.5.2.6.1 Pre-installation coincidence range finder settings. - With the range scale set to 1500 yards or 1500 meters, the calibration reticles positioned in accordance with figure 8 and a 1500 yard or 1500 meters target image in coincidence, the following requirements shall be met:

a. ICS Knob - The setting of the Internal Correction Standard (ICS) knob shall be within 13 Units of Air (UOE's) (graduation) of the mid-position "(25)" and recorded.
b. Horizontal Calibration Knob - The setting of the horizontal calibration knob shall be within 40 (UOE's) (graduation) of the mid-point of total travel.
c. Vertical Calibration Knob - The setting of the vertical calibrating knob shall be within 180 degrees of total travel.

3.5.2.6.2 Final installation calibration settings (coincidence). - With the coincidence range finder installed in a completed vehicle, the coincidence range finder settings (see 3.5.2.6.1) shall be checked to a medium range target.

3.5.2.7 Stero range finder.

3.5.2.7.1 Stero range finder collimation. - The left and right lines of sight of the rangefinder as installed in the turret shall not diverge more than 30 minutes of arc horizontally nor more than 15 minutes of arc vertically, and shall not converge horizontally. These conditions shall be met with the interpupillary knob set at all positions between 58 MM and 72 MM, with and without filters.

3.5.2.8 Accuracy elevation quadrant M13 and M13A1 . - Elevation position of gun shall correspond to reading on Elevation Quadrant M13 and M13A1 within plus or minus 1 mil.

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3.5.2.9 Gun tube and sighting system boresight retention . - The line of sight of the range finder and gunner's periscope shall maintain boresight within plus or minus 0.25 mil in elevation and azimuth, with respect to the axis of the bore of the gun. The gunner's telescope shall maintain boresight within plus or minus 0.15 mil.

3.5.2.10 Superelevation. - Superelevation as measured at the gun shall be within the tolerances specified in table IV when data is transmitted automatically as a function of range scale setting on the range finder.

Table IV. - Superelevation tolerances

Range *Yards or Meters	Tolerance - Mils Plus or Minus
500	0.3*
1000	0.3
1500	0.3
2000	0.4
3000	0.5
4000	0.7

*NOTE: Which ever linear system is employed the figures indicated should be used. For ammunition HVAT, M332 Cam Part Number 8201957 only, the tolerance at 500 yards or meters shall be plus 0.1 and minus 0.5 mils.

3.5.3 Cupola machine gun.

3.5.3.1 Synchronization. - The periscope sight shall be synchronized with the cupola machine gun so that the line of sight of the periscope maintains a given azimuth and elevation relationship (established at horizontal gun position) with the centerline of the machine gun bore within 1.8 mils at all gun positions between 10 degrees depression and 30 degrees elevation and within 3 mils at all gun positions between 30 degrees elevation and maximum elevation.

3.5.3.2 Range of elevation. - The machine gun shall be capable of a minimum of 54 degrees elevation and 10 degrees depression.

3.5.3.3 Blacklash. - Elevation backlash of the cupola machine gun sighting system shall not exceed 0.3 mils at any position.

3.6 Radio suppression. - Turret and gun control and gun fire control system shall be radio interference suppressed in accordance with tactical vehicle requirements of Specification MIL-S-10379.

3.7 Painting, marking and data plates.

3.7.1 Painting. - The exterior and interior of turret, components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with Standard MIL-STD-193. Turret exterior paint and non-skid paint shall match color chip X24087 of Standard FEDERAL STD NO. 595.

3.7.2 Identification and data plates. - Identification and data plates shall conform to Specification MIL-P-514.

3.8 Workmanship. - Workmanship shall be of a quality to assure the hydraulic system free of any foreign matter and to assure that turret and systems are free of defects resulting from improper manufacturing or assembly practices, and shall conform to requirements and tests herein specified.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's inspection. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. - Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced therein.

4.2 Government verification. - All quality assurance operations performed by the contractor will be subject to government verification at unscheduled intervals. Verification will consist of (a) Surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.3 Preproduction sample. - A preproduction sample with simulated payload, shall be submitted for inspection and tests as specified in 3.1. Inspection shall be performed at the place of manufacture. Failure of the preproduction sample to comply with any of the requirements specified, shall be cause for refusal of the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.4 First production turret. - The first complete production turret submitted shall be subjected to examination and tests to determine conformance with specified requirements. The first complete production turret, when installed in the vehicle shall be subjected to the 2000 mile test is specified in table V. Inspection and tests shall be performed by the Government and shall be conducted at a site approved by the Government and will require a maximum of 90 days. Delays caused by the turret and components breakdowns and failure of the contractor to adequately support the turret and components with parts during the tests shall not be the responsibility of the Government. All test mileages indicated in table V shall be accumulative subsequent to 50 miles of break-in operation.

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Table V. - 2000 Mile Test

COURSE	MILEAGE AND SPEEDS	VEHICLE LOADING
Hard Surface	50 mile break-in up to 50% of rated speed	Simulated combat payload
Hard Surface	250 miles at varying speeds up to maximum --	Simulated combat payload
Level Cross Country	850 miles varying speeds	Simulated combat payload
Hilly Cross Country	850 miles at varying speeds	Simulated combat payload

4.4.1 Test failure. - Failure of the first production turret to comply with any of the requirements specified or any major deficiency of a workmanship or materials nature found on the turret during and as a result of the 2000 mile test, shall be cause for rejection of the turret and refusal to continue acceptance of production turrets until objective evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency.

4.5 Materials. - To determine conformance to 3.3, the contractor shall provide the Government with objective evidence that materials used are as specified in the applicable specifications and drawings, and have been tested in accordance with the Quality Assurance Provisions as contained therein.

4.5.1 Qualified products. - To determine conformance to 3.3.1, the contractor shall provide the Government with a certification stating that qualified products, assemblies, and component parts meet requirements of the applicable specifications, and that samples have been tested in accordance with the quality assurance provisions described therein. The certificate must list the tests performed, and the results obtained, and shall be signed by a responsible officer of the certifying agency.

4.6 Construction. - To determine conformance to 3.4, assemblies and components shall be inspected using appropriate drawings to insure that dimensional and process controls are employed in vehicle construction.

4.7 Failure to comply. - Failure of the contractor to comply with inspection provisions 4.5, 4.5.1 and 4.6, of this specification, shall be cause for refusal of the Government to accept any vehicles under the contract as may be submitted by the manufacturer.

4.8 Inspection provisions.

4.8.1 Lot size. - A lot shall consist of all complete assemblies of one type from an identifiable production period submitted at one time for inspection.

4.8.2 Sampling for inspection. - For the purpose of visual, dimensional and primary functional inspection, a representative sample shall be selected from each lot in accordance with inspection level I of Standard MIL-STD-105, using Table VI classification of defects and acceptable quality level:

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Table VI. - Classification of defects

	Characteristics	Major AQL = 40/100	Method of Inspection
101.	Gun balance	Non-conformance	Visual & Functional
102.	Manual traverse	Inoperative	Functional
103.	Rate of travel	Non-conformance	Visual & Functional
104.	Manual elevation	Inoperative	Functional
105.	Rate of travel	Non-conformance	Visual & Functional
106.	System backlash, traverse	Non-conformance	Visual & Functional
107.	Elevation	Non-conformance	Visual & Functional
108.	Power & manual controls, simultaneous operation	Inoperative	Functional
109.	Stability	Non-conformance	Functional
110.	Override control commander's switch	Inoperative	Functional
111.	Gunner's control	Inoperative	Functional
112.	Turret traverse, speed controls	Malfunction	Functional
113.	Tracking speed	Non-conformance	Functional
114.	Maximum speed	Non-conformance	Functional
115.	Stability (all speeds)	Malfunction	Functional
116.	Gun elevation, speed controls	Malfunction	Functional
117.	Transition (between speeds)	Erratic operation	Functional
118.	Control system, dead spot, gunner's and commander's switch	Inoperative	Functional & Visual
119.	Traverse	Non-conformance	Functional & Visual

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Table VI. - Classification of defects (Cont'd)

120.	Elevation	Non-conformance	Functional & Visual
121.	Elevation system protection	Inoperative	Functional & Visual
122.	Positive stop (elevation)	Non-conformance	Functional & Visual
123.	Positive stop (depression)	Non-conformance	Functional & Visual
124.	Backlash, (sighting system)	Excessive backlash	Functional & Visual
125.	Line of sight, synchronization	Non-conformance	Functional & Visual
126.	Elevation tracking	Non-conformance	Functional & Visual
127.	Boresight knob travel, up and down	Non-conformance	Functional & Visual
128.	Right and left	Non-conformance	Functional & Visual
129.	Leveling devices	Non-conformance	Functional & Visual
130.	Range finder, collimation	Non-conformance	Functional & Visual
131.	Convergence - horizontally	Non-conformance	Functional & Visual
132.	Accurancy, elevation quadrant	Non-conformance	Functional
133.	Super elevation	Non-conformance	Functional

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Table VI. - Classification of defects (Cont'd)

	Characteristics	Minor AQL = 100/100	Method of Inspection
201.	Gun balance	Improper assembly and in stallation	Visual & Functional
202.	Manual traverse	Excessive handcrank effort, excessive chatter	Gage & Functional
203.	Manual elevation	Excessive handcrank effort, excessive chatter	Gage & Functional
204.	Power & manual controls	Malfunction	Functional
205.	Commander's and gunner's override controls	Malfunction	Functional
206.	Gun elevation controls	Faulty operations	Visual & Functional
207.	Dead spot in traverse and elevation control systems	Non-conformance	Visual & Functional
208.	Elevation system protection	Non-conformance	Visual & Functional
209.	Hand traverse and elevation crank travel-movement	Excessive	Visual & Functional
210.	Leveling devices	Faulty adjustment	Visual & Functional
211.	Painting	Improper application and color	Visual
212.	Decals, marking data & instruction plates	Data incomplete and missing, improper location and color	Visual
213.	Workmanship	Excessive rust & weld spatter, loose nuts & bolts, missing cotter pins, safety wires, chains, lock pins, wiring clips, ground strips, & grommets, damaged & missing lube fittings, stripped threads & faulty installations	Visual

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4.9 Turret and gun control performance test. - The turret and gun control system of each vehicle shall be subjected to all tests specified herein at any metal temperature of the turret control parts between minus 25° to plus 125° F. Each test specified herein shall be conducted at the place of manufacture except for extreme ambient temperature testing which shall be performed at a Government proving ground. Unless otherwise specified, all tests shall be performed with the vehicle level as specified and with the gun balanced as specified (see 3.5). Failure, to pass any of the tests specified herein, shall be a basis for rejection.

4.9.1 Turret and gun control system test.

4.9.1.1 Gunner's power control test. - To determine conformance to 3.5.1.1, the gunner's power control shall be operated throughout all ranges of operation and the deadspot and backlash not to exceed specified degrees.

4.9.1.2 Commander's power control test. - To determine conformance to 3.5.1.2, the commander's power control shall be operated throughout all ranges of operation and the function of controls checked.

4.9.1.3 Deck clearance valve test. - To determine conformance to 3.5.1.3, the gun shall be operated over the rear deck, sides of the tank, and super structure and function of the valve checked.

4.9.1.4 Superelevation actuator test. - To determine conformance to 3.5.1.4, with zero superelevation set on the computer, range 4000 yards or 4000 meters for HEP-T-T142E3 ammunition with the Range Servo Switch turned off. Turn the Range Servo Switch on. The gunner's periscope reticle shall be on target within the specified limits after automatic transmission of superelevation.

4.9.1.5 Gun sighting system test. - To determine conformance with 3.5.1.5, the sighting system shall be operated and the synchronization of system checked.

4.9.1.6 Manual traverse effort test. - To determine conformance to 3.5.1.6, the force required shall be measured to rotate the hand traverse crank to maintain movement of the turret. The force shall be measured at four points 90 degrees apart on the turret race assembly and in each direction at all points.

4.9.1.7 Manual elevation effort test. - To determine conformance to 3.5.1.7, the force required shall be measured to turn the manual elevation hand crank to maintain movement of the gun in elevation and depression. The force shall be measured with the gun at horizontal $\pm$ 1 degree, at plus 15 degrees $\pm$ 1 degree elevation and at minus 5 degree $\pm$ 1 degree depression.

4.9.1.8 Manual traverse response rate test. - To determine conformance to 3.5.1.8, the hand traverse crank shall be rotated one or more revolutions and the angular movement of the turret shall be measured by means of the azimuth indicator. Only one check point in 360 degrees shall be required.

4.9.1.9 Manual elevation response rate test. - To determine conformance to 3.5.1.9, the gun angular movement per revolution of the hand crank shall be measured by means of the M13A1 quadrant for at least ten revolutions of the hand crank. The angular movement shall be measured in elevation and depression.

4.9.1.10 System backlash test. - To determine conformance to 3.5.1.10, a force of 70 pounds minimum shall be applied in the plane of rotation perpendicular to a level gun twenty inches from the muzzle end of the main gun. After referencing the position of the gun with the azimuth indicator and M1 gunner's quadrant, the angular movement of the gun shall be measured when the force of 70 pounds is applied in the opposite direction.

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4.9.1.11 Control transfer test. - To determine conformance to 3.5.1.11, the controls shall be operated in any combination and simultaneously with manual and power controls, and with the controls centered, the turret and gun shall be observed for movement.

4.9.1.12 Controlled traverse velocity test. - To determine conformance to 3.5.1.12, with 26 volts applied to the hydraulic pump and gun traversed, the control of the gun shall be checked.

4.9.1.13 Traversing and elevation stability test. - To determine conformance to 3.5.1.13 and 3.5.1.14, the vehicle master switch shall be turned off and the gun movement shall be observed at the end of a 12 hour period. It shall be permissible to reduce the 12 hour period to a 3 hour, 6 hour or 9 hour period provided the allowable gun movement is reduced proportionately. The gun movement for 3 hour period shall not exceed 0.25 mils, for a 6 hour period shall not exceed 0.5 mils, or for 9 hour period shall not exceed 0.75 mils. After the requirement has been met with the vehicle master switch off, the vehicle master switch and the turret control switch shall be turned on and gun movement shall be observed at the end of a 3, 6, 9 or 12 hour period. Allowable gun movement for each period shall have the same requirement as that specified with the vehicle master switch off. The vehicle engine shall not be operated during this test. The turret shall be mounted on a turret stand or to a hull when checking this requirement.

4.9.1.14 System protection test. - To determine conformance to 3.4.1.15, the gun shall be driven once against the elevation and depression stops first by power and then by manual control at full speed. Any time delay in regaining power or manual control is basis for rejection.

4.9.1.15 Manual elevation supercharge test. - To determine conformance to 3.5.1.16, test the manual elevation hydraulic system by raising or lowering the gun by means of the hand pump after the system has been idle 48 hours.

4.9.1.16 Elevation and depression limit test. - To determine conformance to 3.5.1.17, the gun elevation and depression angles shall be measured in each of the defined areas when using power and manual controls.

4.9.1.17 Gun elevation, speed and depression tests. - To determine conformance to 3.5.1.18, the gun shall be elevated and depressed at all speeds between 0.5 mil per second and 4 degrees per second.

4.9.1.18 Turret friction test. - To determine conformance to 3.5.1.19, with the tank on level terrain the torque required to maintain turret rotation shall be not more than specified foot pounds, and variation between maximum and minimum shall be not more than specified foot pounds.

4.9.1.19 Operation on slope test. - To determine conformance to 3.5.1.20, with vehicle pitched or on specified slopes the gun shall be operated throughout all ranges of elevation, depression, and traverse, and the function of gun observed.

4.9.1.20 Travel test. - To determine conformance to 3.5.1.21, with the vehicle being operated as specified in the 8 mile road test of the vehicle Specification MIL-T-45148, the elevation and traverse hand-crank shall not evidence any spinning movement.

4.9.1.21 Time required to lay gun on stationary target test. - To determine conformance to 3.5.1.22, the gunsight reticle shall be brought within the borders of the target specified, for each condition in the average time specified in table I.

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4.9.1.22 Tracking velocity test. - To determine conformance to 3.5.1.23, with a minimum of 26 volts potential applied to the input of hydraulic power unit, the gun shall be operated and the performance observed.

4.9.1.23 Tracking accuracy test. - To determine conformance to 3.5.1.24, the moving target shall be tracked by the gunner and commander power controls while observation is made through the sighting system of the percentage of time the reticle remains within the specified limits.

4.9.1.24 Test for gun laying on a moving target . - To determine conformance to 3.5.1.25, the time shall be measured to bring the gun sight reticle within the borders of a target as specified in table III. The lead angle shall be established from a fixed aiming point behind the target. The fixed aiming point shall be a stationary or moving point relative to the target. After the reticle has been positioned within the borders of the target, the reticle shall remain within the target borders for the completion of the run in accordance with percentage of time on target specified in table II. The requirements shall be met with gunner and commander power controls.

4.9.2 Gun fire control system test. - After installation, all instruments and devices shall be synchronized as specified in 3.5.2.2, and tested as specified in 4.9.3 before further testing and inspection. Figures 1 through 8 cover details for testing of the system.

4.9.3 Synchronization test. - To determine conformance to 3.5.2.2, the line of sight of optical components shall be tested by the method specified in 4.9.3.1, or if approved by the Government inspector, by the method in 4.9.3.2.

4.9.3.1 Turret stand and fixed target method .

4.9.3.1.1 Equipment. - The gun sighting system shall be inspected by using the following equipment:

(a) Turret stand. - The turret stand shall be sufficiently rigid to withstand the stresses of normal testing procedures without distortion or variation in the turret mounting plane. The turret stand, with turret in place, shall maintain the distance from the center of the gun trunnions to the plumb line plane through the horizontal target aiming point within plus or minus 1/8 inch of target calculated distance. With the turret mounted on the stand, the minimum distance between the gun trunnions and the target shall be 40 feet.

(b) Target. - The target shall be a rigid structure made of steel with the aiming points located to reflect 1500 yards or 1500 meters boresight. The surface of each aiming point shall not deviate from plumb line plane that passes through the horizontal gun-aiming point. The vertical plane through the surface of the aiming points shall be perpendicular to the plane of gun elevation within plus or minus 3 mils. Gun and sighting instrument aiming points shall be provided at zero elevation, 15 degrees elevation and at two positions approximately equally spaced between these limits. Gunner's telescope aiming points are not required at elevated positions because the rigid mounting of the telescope directly to the gun mount and shield

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insures synchronization. Only a left line of sight series of aiming points for the range finder shall be provided at positions above zero elevation. Sighting instrument aiming points shall incorporate a correction for sight location parallax error only. The target shall be capable of rotating a few degrees both clockwise and counterclockwise and may be adjustable above and below the horizontal gun position. The target shall be illuminated to the extent that the aiming points are adequately visible when aperture covers are placed on the sighting instruments.

(c) Telescope. - The telescope shall be an observation telescope TT and H 112 or 392, 20-power focus for reticle and field, adjustable independently of one another, or approved equal. The telescope shall be mounted in a self-alining type mount which will center the telescope in the gun tube.

(d) Aperture plugs. - Aperture plugs shall be provided to define the target and to minimize parallax on the range finder, and gunner's periscope. The plugs shall attach firmly to the various objective windows and shall position the aperture in the center of the window. Aperture diameters shall be 1/8-inch diameter, maximum, to eliminate parallax error.

4.9.3.1.2 Checks and adjustments. - With all gun sight instruments installed and the turret assembly on the turret stand, necessary adjustments and checks shall be made in accordance to the following procedure:

(a) Telescope and mount assembly shall be inserted in the gun tube chamber.
(b) The target shall be positioned with respect to the gun trunnions to conform to the following conditions.

1. When the gun is leveled by laying low to high with the gunner's quadrant M1 on the gun breech ring pads the target horizontal gun aiming point shall be in line with the crosshair reticle in the telescope T. T. and H. 112 or 392 within 1/8 inch. The target may be raised or lowered to position the aiming point.
2. The line of sight of the gun shall exactly follow the line through the gun aiming points on the target. The target shall be rotated no more than 2 mils clockwise or counterclockwise from a plumb line position to locate the target.

(c) The ballistic computer superelevation counter shall be positioned at zero by rotating the hand crank.
(d) Aperture plugs shall be placed over the objective windows of the range finder, and gunner's periscope.
(e) The range finder shall be set to 1500 yards or 1500 meters, and the left and right side aiming points superimposed by means of the horizontal and vertical calibration knobs.

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(f) Reticles of the sights in (d) above shall be alined with corresponding aiming points by rotating the azimuth and elevation bore sight knobs.

4.9.3.1.3 Procedure. - The boresight locking levers shall be applied and the boresight knobs shall not move more than 0.05 mils when the locking levers are applied. The gun shall be positioned at each of the aiming points by laying low to high and telescope cross hairs shall fall within their respective tolerance bands on the target. A reticle line shall be considered within the tolerance bands if half or more of the reticle line width is in the white area. All reticle cross hairs shall be exactly alined with their corresponding aiming points when the gun is positioned on the horizontal aiming points by laying low to high. Should a progressive error be indicated during inspection of the sighting system, the controlling linkages shall be adjusted accordingly and items after (e) above repeated. Adjustment of linkages to correct deviation of the line of sight of any specific sight from vertical plane shall not exceed a maximum of 15 minutes of arc (3 graduations) from level. In the event of the need for an adjustment of more than 5 minutes of arc, the source of error shall be determined and corrected prior to acceptance.

4.9.3.2 Aiming point method (point of infinity).

4.9.3.2.1 Equipment. - The equipment specified below shall be used in determining synchronization of the gun sighting instruments when the inspection is made after installation in the vehicle.

(a) Ramp or hill with a slope of approximately 15 degrees.
(b) Binoculars, seven power or better.
(c) Twine or wire for use on end of gun muzzle as cross hairs.
(d) Quadrant, gunner's M1.

4.9.3.2.2 Procedure. - The following procedure shall be followed:

(a) Twine or wire shall be installed as cross hairs on muzzle of gun.
(b) Percussion mechanism shall be removed from breech block.
(c) The ballistic computer superelevation counter shall be positioned at zero by rotating the hand crank.
(d) Place the right telescope of the binocular over the firing pin well. Using this as a rear sight and the crosshairs on the end of the gun muzzle as a front sight, the gun shall be laid accurately from low to high on a target at a range of not less than 1500 yards or 1500 meters.
(e) The gunner's and commander's sights shall be laid by using elevation and azimuth boresight knobs at the same point as in (d) above. Boresight knobs shall be locked and graduated scales slipped to normal position.
(f) Gunner's quadrant shall be set at 267 mils and placed on the breech ring of the gun.
(g) The tank shall be driven up the ramp until the bubble in the gunner's quadrant is approximately level. The tank brakes shall be applied.

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(h) Gun shall be re-laid on the same target as in (e) above. The gunner's and commander's periscopes shall be checked to determine if their respective gun aiming reticles are now aiming at the same point as the gun. If not, the boresight knobs shall be unclamped and the reticles moved until they are on target. Movement of boresight knobs required to bring the reticle on target of less than 0.5 mils shall indicate synchronization of gunner's and commander's sights within permissible tolerance.

4.9.4 Leveling devices. - The gun shall be leveled at zero elevation as read by use of the gunner's quardant M1 positioned on the breech ring pads. The leveling devices on the range finder and ballistic drive shall be observed to ascertain conformance to 3.5.2.5.

4.9.5 Boresight knob travel test. - All boresight knobs shall be checked for conformance with 3.5.2.4. The following conditions shall apply:

(a) Superelevation counter of the ballistic computer shall be positioned at zero position. The ballistic drive superelevation input shaft will not be qualified.
(b) The range finder range scale shall be set to 1500 yards or 1500 meters, and the aiming points superimposed by means of the horizontal and vertical calibration knobs.
(c) The line of sight of the range finder (left and right), gunner's periscope and gunner's telescope shall be alined accurately on their corresponding aiming points.

4.9.6 Backlash . - Sighting system backlash shall be checked after satisfactory completion of synchronization inspection to determine conformance to 3.5.2.1. Checks shall be made and results noted in accordance with the following procedure:

(a) Place gun "dead on" aiming point of target by laying from low to high. This shall be accomplished without over travel.
(b) Place all sighting equipment gun aiming reticles "dead on" target relative to the gun. Lock elevation and azimuth boresight knobs in position of direct alinement of gun aiming reticles on target aiming point. Slip micrometer scales to nominal setting.
(c) The gun shall be realined with the same target aiming point as in "a" by laying from high to low without over travel.
(d) If necessary, realine sight reticles "dead on" aiming point by use of elevation boresight knobs. Read elevation boresight knob micrometer index scale.
(e) The difference in reading between "b" and "d" above must not exceed the specified backlash defined in 3.5.2.4.

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(f) Backlash shall be checked in both directions, this will be accomplished by reversing directions in "a" and "c" then repeating paragraphs "a" thru "e". Backlash should also be checked at angles of elevation above zero to attain a more precise determination of inherent backlash effects.

4.9.7 Superelevation. - Superelevation in the gun sighting system shall be checked according to the following procedure and shall be observed to determine conformance to 3.5.2.10.

(a) Ballistic computer range correction knob shall be positioned at zero.
(b) If a mil scale on the target is to be used to read gun superelevation, the gun shall be alined with the horizontal gun aiming point by laying low to high. The mil scale shall be fixed to the target with the zero index at the horizontal gun aiming point. If the gunner's quadrant M1 is to be used to measure gun superelevation, the gun shall be leveled as read on the gunner's quadrant M1 positioned on the breech ring pads.
(c) The gunner's periscope shall be alined with its aiming point by rotating the boresight knobs. The knobs shall be locked.
(d) The range finder range scale light shall be switched on.
(e) The range finder range scale shall be positioned at 500 yards or 500 meters by rotating the ranging knob from a position below 500 yards or 500 meters without over-traveling.
(f) The ballistic computer switch at the range finder shall be turned on.
(g) The gunner's periscope shall be lowered as a result of superelevation input from the ballistic computer. The gunner's periscope reticle shall be realined with its aiming point by elevating the gun. Alinement shall be accomplished from low to high.
(h) Superelevation of the gun shall be noted by the position of the gun tube telescope reticle in the mil target or by reading the gunner's quadrant M1 on the gun breech ring pads. Superelevation readings shall conform to table VII with tolerances as specified in 3.5.2.10. The ballistic computer power switch at the range finder shall be turned off.
(i) The inspection procedure in (a) through (h) above shall be repeated at the ranges in table VII.
(j) Gun superelevation tests as defined herein shall be performed on those ammunitions indicated in the ballistic computer ammo indicator window.

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Table VII. - Superelevation (Mils.)

AMMO	Cam Part No.	yds 500	meters 500	yds 1000	meters 1000	yds 1500	meters 1500	yds 2000	meters 2000	yds 3000	meters 3000	yds 4000	meters 4000
APT M318A1													
HE M71													
HEAT T300E56													
WP M313													

4.9.8 Test for elevation quadrant M13 and M13A1. - With the turret in an approximately level position, using the M1 calibrated gunner's quadrant on the breech pads, elevate the gun to 19 degrees (337.78 mils). Check reading of elevation quadrant M13 and M13A1 to determine conformance to 3.5.2.7. The procedure shall be repeated with the gun at a depression of 8 degrees (142.22 mils).

4.9.9 Coincidence range finder calibration checks.

4.9.9.1 Pre-installation check. - The range finder shall be mounted in a range finder final inspection fixture and checked to determine conformance to 3.5.2.6.1.

4.9.9.2 Turret installation check on vehicle. - With the range finder mounted in the turret, range finder calibration shall be checked in accordance with the following procedure:

a. Provide a clearly defined target between 1380 yards to 1620 yards located within an accuracy of 1/3 of 1 percent and set the range scale to agree with the exact range chosen.
b. Set the occuler knob to its mid-position.
c. With the eyepiece in proper focus for the observer, establish coincidence with the target by means of the horizontal and vertical calibrating knobs.
d. Illuminate the calibrating reticle (the calibrating reticle lamp shall not be left on for more than 10 seconds at any one time; if it is necessary to turn on the reticle lamps again, allow at least one minute between checks) establish alinment of the coincidence reticle (See Figure 8) using the cover protected ICS and halving knobs.
e. Check for conformance to paragraph 3.5.2.6.1, a, b, and c.
f. If the target cannot be brought into coincidence, the vehicle shall be considered unacceptable until the cause of misalinment has been determined and remedied.

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4.9.10 Stereo range finder collimation test. - With range finder mounted in the turret, divergance shall be checked according to the following procedure or with an approved collimation fixture and target to determine conformance to 3.5.2.7.

(a) The range finder shall be sighted on a target at a distance of 650 to 4800 yards or meters. The target shall be located with an accuracy of 1/3 of 1 percent.
(b) The double collimator support fixture shall be installed on the range finder.
(c) Range knob shall be set to position the range scale at the target distance.
(d) The double collimator shall be placed on the eye pieces.
(e) With the eye pieces focused, the line of sight shall be checked to rmine conformance to 3.5.2.7.

4.9.11 Boresight retention test . - After the eight mile road test specified in 4.9.1.20, boresight retention tests shall be performed on inspection control vehicles at Government proving ground facilities as specified below to determine conformance to 3.5.2.9.

4.9.11.1 Boresighting. - Prior to the road test (see 4.9.1.20) the line of sight of the range finder, gunner's periscope, and gunner's telescope shall be alined accurately with the centerline of the gun tube.

4.9.11.2 Boresight check.- After the eight mile road test (see 4.9.1.20), a check of boresight retention shall be made as follows:

(a) The equipment as specified in 4.9.3.2.1 shall be used to lay the gun on the same target used in 4.9.11.1.
(b) The azimuth and elevation boresight knobs on the gunner's periscope shall be used, if necessary, to position the periscope's line of sight dead on target.
(c) The amount of mil travel on the periscope's boresight knobs shall be observed to determine conformance to 3.5.2.9.
(d) The steps (a) through (c) shall be repeated for the range finder which has been recalibrated to establish coincidence of the target.
(e) The steps (a) through (c) shall be repeated for the range finder.
(f) With the gunner's periscope boresighted on target, the reticle of the gunner's telescope shall be laid on the same target using the gun elevating and turret traversing controls.
(g) The gunner's periscope reticle shall be laid again on target with the periscope boresight knobs, noting the mil travel on the knobs.
(h) The readings obtained in (f) shall be the shift in boresight of the gunner's telescope, but the direction of shift shall be reversed.
(i) Through this operation, the gunner's telescope shall remain in locked position. The boresight knobs on the gunner's telescope shall not be used for checking shift in boresight.

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4.9.11.3 Repeat of boresight retention test. - If any boresight knob has to be moved more than the values specified in 3.5.2.9 to bring the line of sight of the respective sighting instrument "dead-on" the target, all instruments shall be reboresighted in 4.9.11.1, the ten mile road test repeated, and a second boresight check made as in 4.9.11.3. If any instrument will not maintain boresight as specified in 3.5.2.9 after the second ten mile road test, the vehicle shall be considered unacceptable until the cause for the loss of boresight is determined and remedied.

4.9.12 Mount test. - The gun shall be elevated and depressed to its maximum positions and performance observed to determine conformance to 3.5.2.3.

4.9.13 General inspection of turret and gun control and gun fire control system. - After completion of synchronization and inspection thereof, general inspection of the control system shall include at least the following items:

(a) All periscope heads shall be removed and replaced to assure that it is possible to do so without interference.
(b) The elevation scan lever shall be snapped on each periscope head to insure the "seating" of the coupling.
(c) Clearance around bullet splash guards shall be measured to assure minimum clearance of not more than 1/4 inch.
(d) Seals around periscope heads and the range finder shall be examined for proper installation.
(e) All exposed glass surfaces shall be examined for signs of mars such as chips and scratches.
(f) Headrests shall be examined for freedom of dirt and oil.
(g) Electrical connections and controls for illumination of all scales, reticles, etc., and circuit controlling signal lights shall be inspected for proper connections and operation.
(h) All mounting bolts, including stud connection on cradles, shall be inspected for tightness.
(i) Gun travel in elevation shall be observed to ascertain that the stops are set to specified limits and that the gun sighting instruments are not acting as the gun stop.
(j) Paint finishes shall be inspected and gunsight knobs, scales, eyepieces, firing charts, etc. shall be examined for improper painting.

4.10 Cupola machine gun sighting system.

4.10.1 Equipment. - The equipment specified below shall be used in checking the cupola machine gun sighting system.

(a) Target. - The target shall have aiming points at minus 10 degrees, 0 degrees, plus 15 degrees, plus 30 degrees, and plus 50 degrees. Aiming points shall be in accordance with figure 6 or its substantial equal. The target shall be illuminated so that all aiming points have a minimum brightness of 100 foot-lamberts.
(b) Boresight telescope (with adapter as required) . - The adapter shall have leveling pads parallel to the centerline of the machine gun bore within 0.05 mil.

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(c) M1 quadrant (with calibration accuracy sheet).
(d) Calibrated periscope sight. - A calibrated Periscope Sight shall be installed in the Periscope Mount and shall be used during testing specified in 4.10.2. Sight shall have an input linkage arm assembly with a functional length of 2.250 plus or minus 0.001 inches, and parallax on the optical axis shall not exceed 0.25 mils when viewing a target located between 132 inches and 205 inches from the periscope. The eyepiece shall be adjusted to zero diopter position. A 3x to 6x telescope shall be used to observe through the sight.

4.10.2 Procedure. - The cupola machine gun sighting system shall be inspected, using the procedure specified herein.

4.10.2.1 Synchronization test. - Aline the bore of the machine gun with the center of its horizontal aiming point by laying from low to high. Adjust the periscope sight until its line of sight is coincident with the horizontal aiming point. While laying in as directed above check the ability of the machine gun and periscope sight to meet the synchronization requirements of 3.5.3.1.

4.10.2.2 Machine gun elevation test. - The elevation stop for the machine gun shall be set to allow the maximum elevation through full traverse of the cupola without interference of the machine gun and range finder. This shall be checked with the 90MM. gun at full depression and the range finder at maximum superelevation. The machine gun and periscope sight shall be tested to determine conformance to 3.5.3.2 by depressing the machine gun until the centerline of its bore coincides with the aiming point at 10 degrees depression and by elevating the gun until its maximum elevation is obtained. The gun's maximum elevation shall be measured by using the calibrated M1 quadrant.

4.10.2.3 Backlash inspection. - The machine gun bore centerline shall be alined with its horizontal aiming point by laying from low to high. The periscope sighting reticle shall be adjusted to coincide with its aiming point. The gun shall be re-alined on its aiming point by laying from high to low. The amount which the periscope line of sight lags with respect to its aiming point is the backlash. Repeat the above procedure at 50 degrees elevation. Backlash at the horizontal, and 50 degree elevation check point shall meet the requirements specified in 3.5.3.3.

4.11 Control tests.

4.11.1 Frequency. - To determine conformance to 3.5 through 3.5.2.9, the complete turret and fire control system, identified as to production period, shall be selected at the rate of 1 per month or 2 from each 100 produced, except that not more than 4 shall be selected in any given 30 day period.

4.11.2 Tests, turret and fire control. - The complete turret and fire control system selected in accordance with 4.8 shall be subject to tests specified in 4.8 through 4.10.2.3.

4.11.3 Test failure. - If a turret and fire control system fails to pass any of tests specified herein, the Government inspector shall stop acceptance of subsequent lots until objective evidence has been provided by the contractor that corrective action has been taken.

4.11.4 Proving ground test _____. - One turret and fire control system installed on vehicle from the first 10 submitted for acceptance, and thereafter additional turrets and fire control systems in accordance with a schedule specified by the procuring activity shall be selected by the Government inspector for proving.

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- 4.12 Parts and components. - Parts, components and assemblies shall be inspected in accordance with Supplementary Quality Assurance Provisions (SQAP's) Index 8750104 and 8750105.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage shall be in accordance with Specification MIL-T-45052.

6. NOTES

6.1 Intended use. - The turret, gun, and the fire control systems are intended for use in the M48A2 and M48A2C tank by the Armed Forces as a combat vehicle.

6.2 Procurement document should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction model if required (see 3.1).

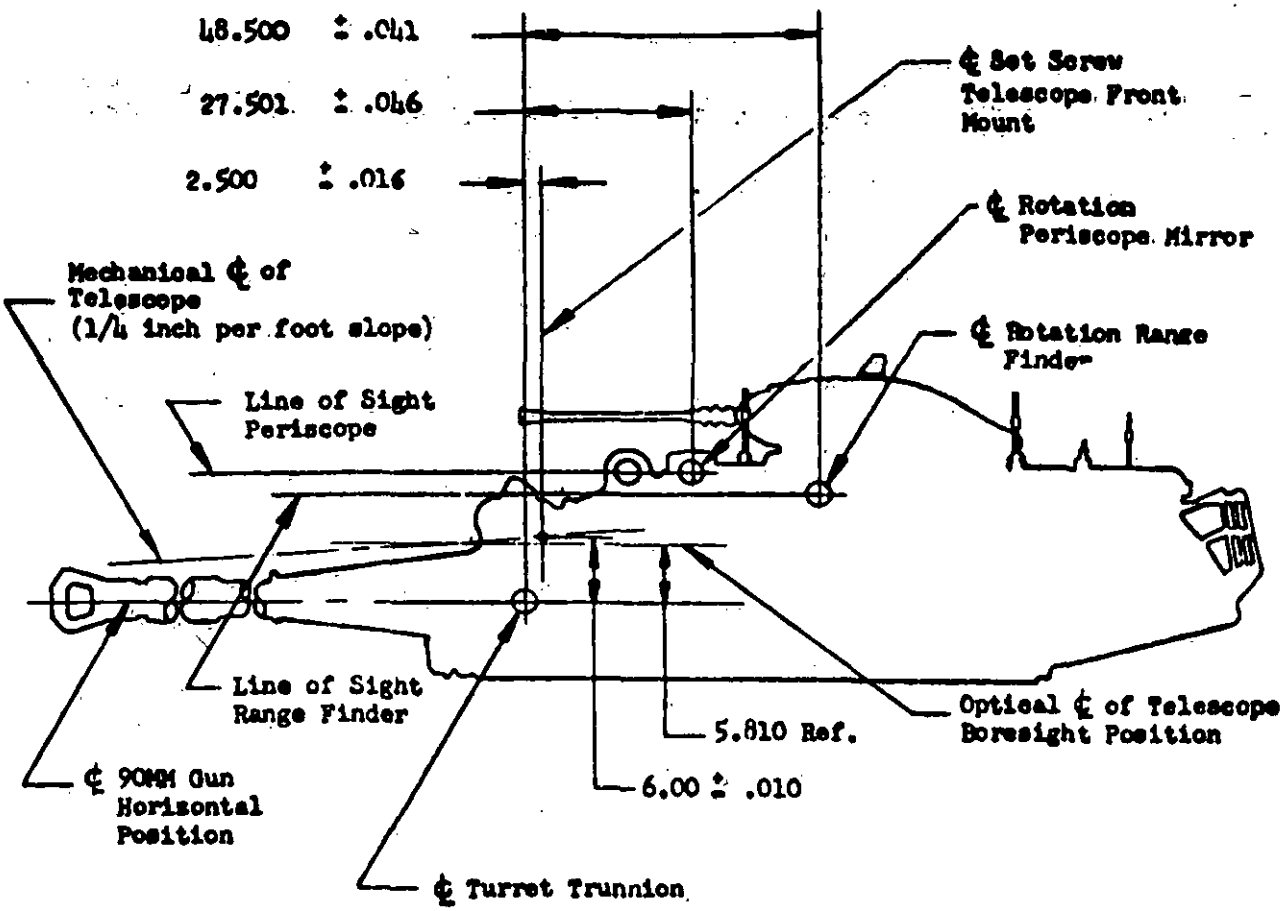
6.3 Safety precautions. - Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army - Ord

Preparing activity:
Army - Ord

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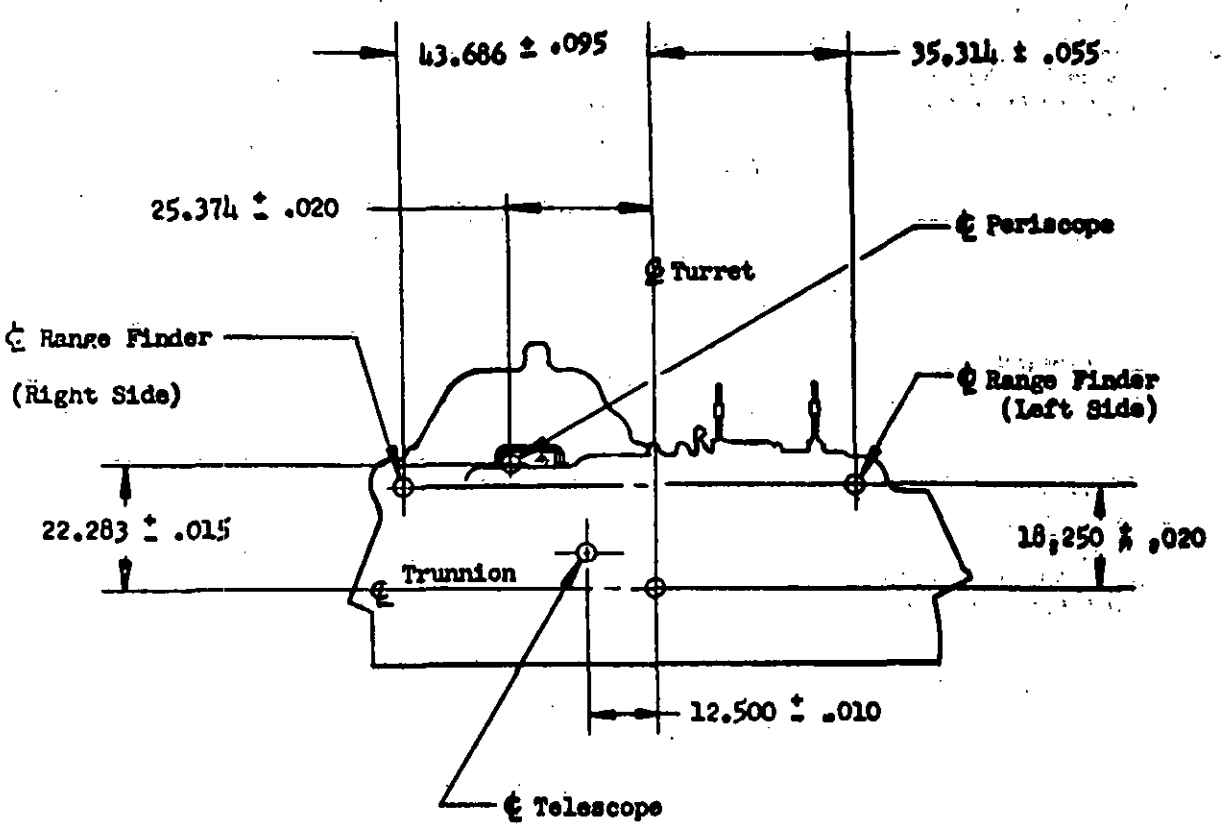


SIDE ELEVATION

NOTE: Maximum Tolerance Accumulation is shown with each sight location dimension.

Figure 1

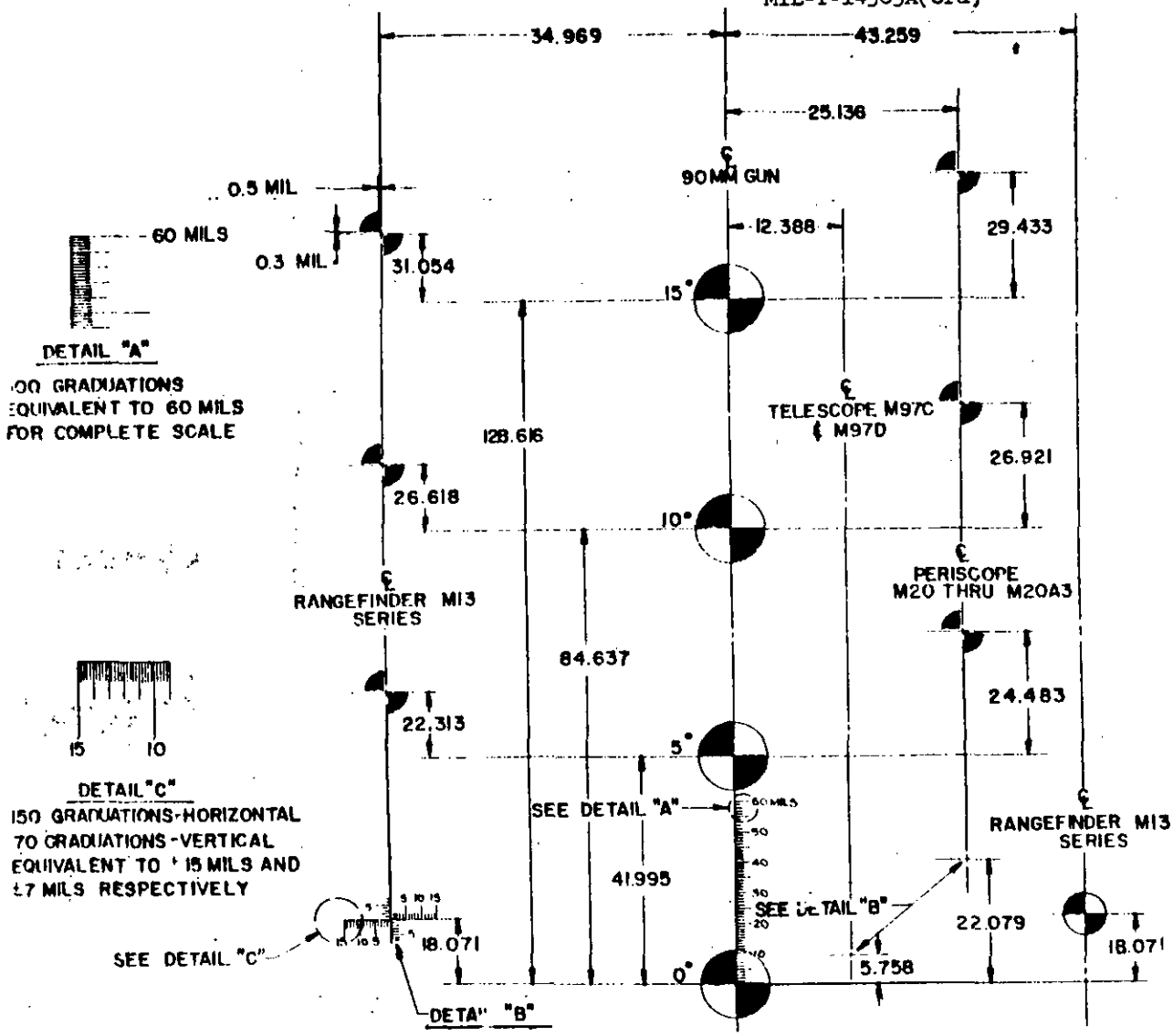
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FRONT ELEVATION

Figure 2

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NOTES

DISTANCE FROM TARGET TO ℄ GUN TRUNNION- 40 FEET

±.005 TOLERANCE ON ALL DIMENSIONS (ALL DIMENSIONS IN INCHES UNLESS OTHERWISE STATED)

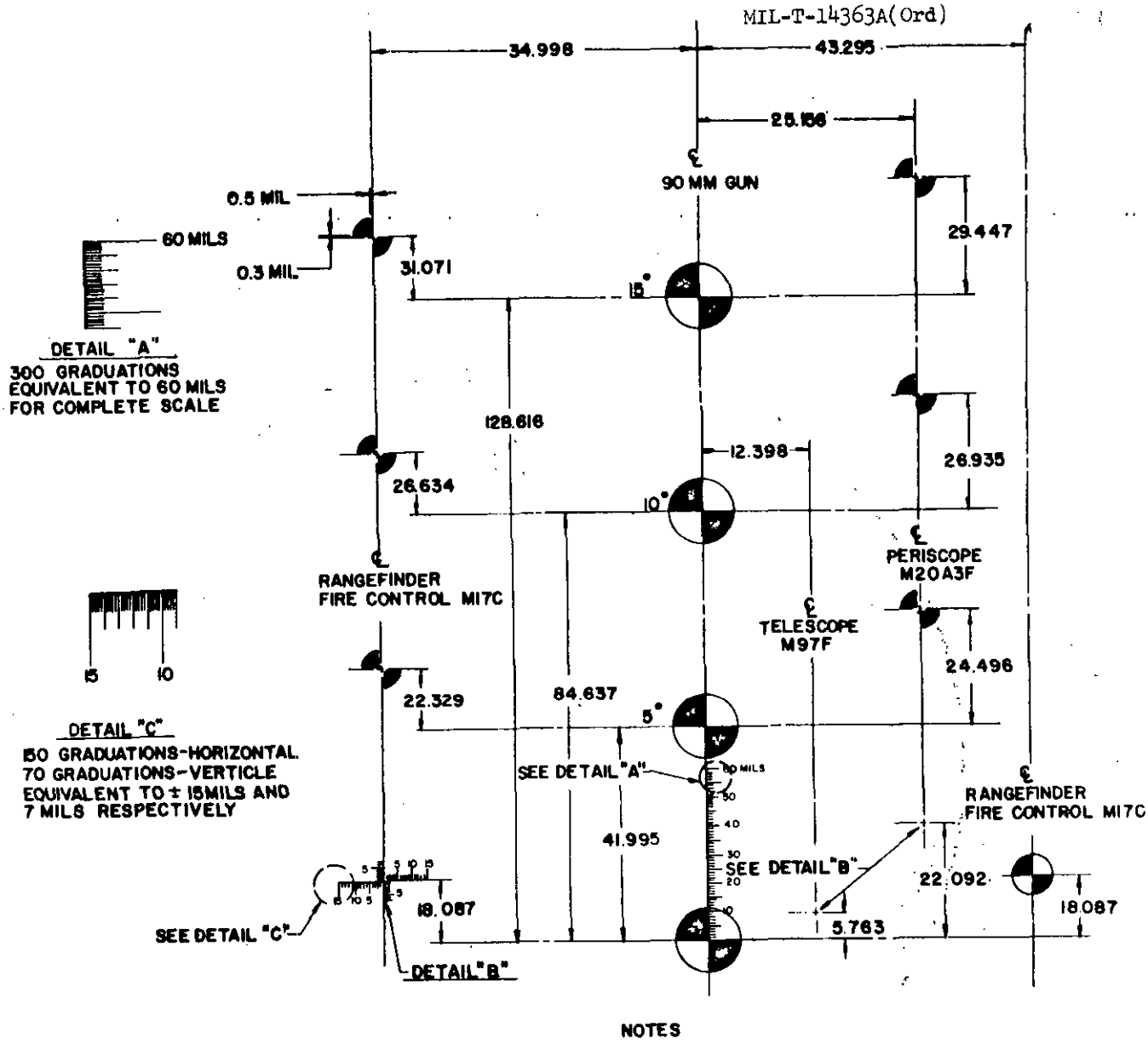
TARGET AIMING POINTS ARE POSITIONED TO REFLECT CONVERGENCE OF THE LINES OF SIGHT WITH THE MAIN GUN AT 1500 YARDS AND ARE CORRECTED FOR SIGHT LOCATION PARALLAX

FIGURE 3 INDOOR TARGET FOR BORESIGHT AND SYNCHRONIZATION TARGET

YARD SYSTEM

See Fig. 3A Metric System

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NOTES

DISTANCE FROM TARGET TO ℄ GUN TRUNNION = 40 FEET

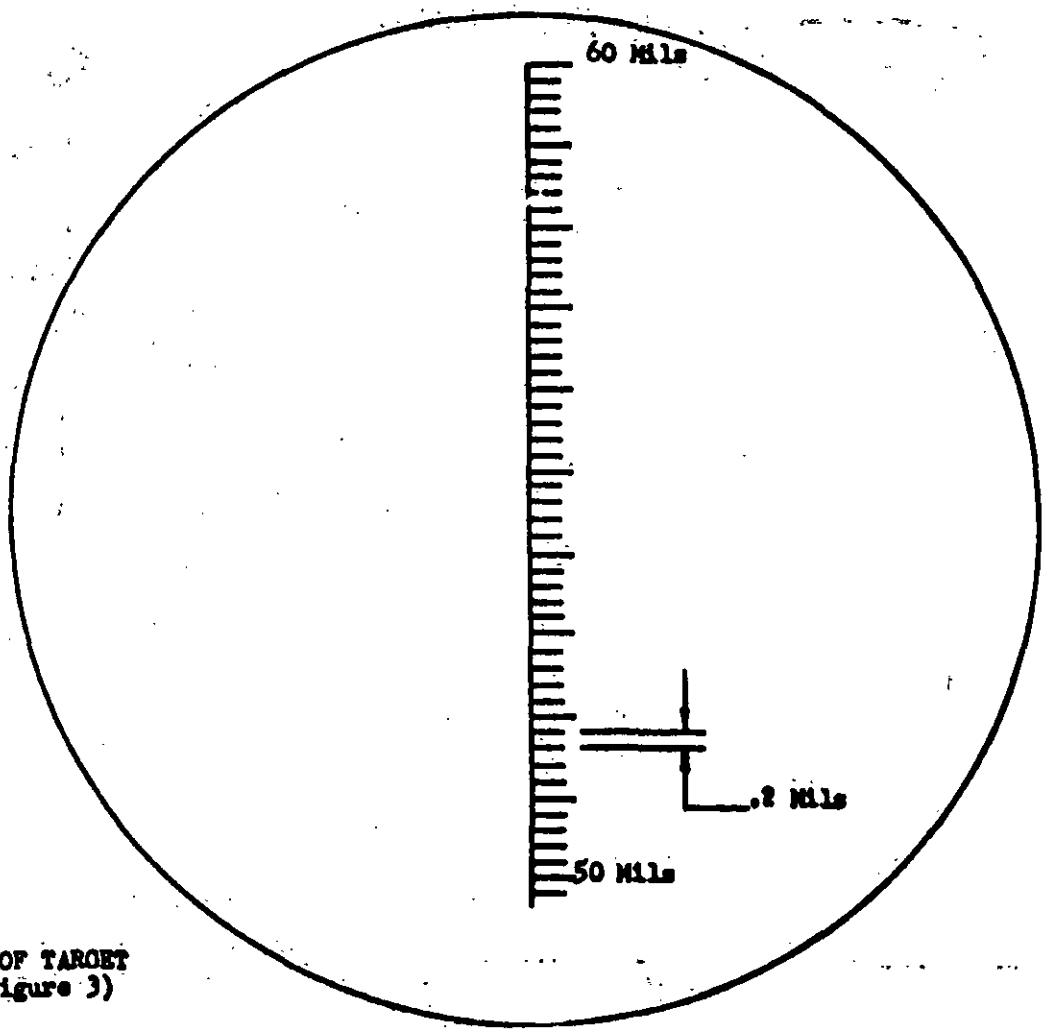
±.005 TOLERANCE ON ALL DIMENSIONS (ALL DIMENSIONS IN INCHES UNLESS OTHERWISE STATED)

TARGET AIMING POINTS ARE POSITIONED TO REFLECT CONVERGENCE OF THE LINES OF SIGHT WITH THE MAIN GUN AT 1500 METERS AND ARE CORRECTED FOR SIGHT LOCATION PARALLAX

Figure 3A **Indoor Target for** BORESIGHT AND SYNCHRONIZATION TARGET

Metric System
See Fig. 3 Yard System

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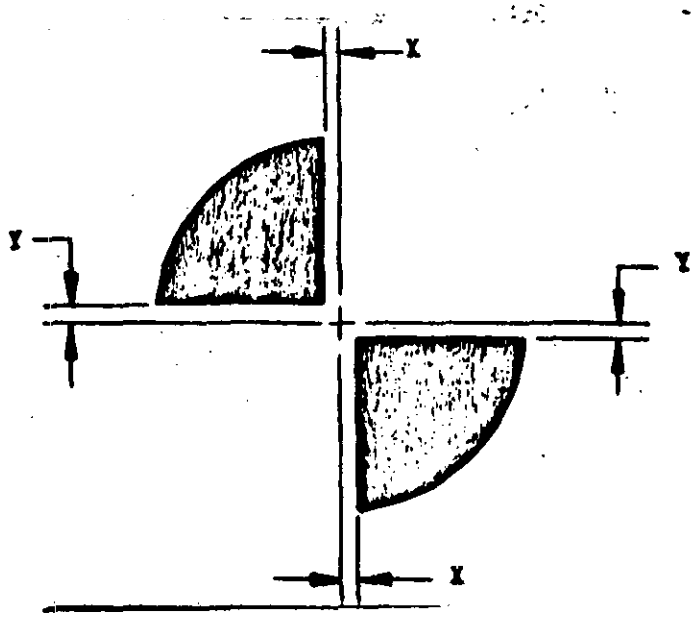


DETAIL OF TARGET
(See Figure 3)

Figure 4

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OPEN AIMING POINTS



0.5 MIL AZIMUTH 0.3 MIL ELEVATION TOLERANCE
PROJECTED ON 40 FOOT TARGET, INCHES

		GUN ELEVATION, DEGREES			
		0	5	10	15
Range Finder	M13A1 M17C				
	X •	.259	.260	.263	.269
	Y •	.156	.157	.161	.167
Periscope	M20A3 M20A3F				
	X •	.249	.250	.253	.258
	Y •	.150	.151	.154	.160

Figure 5 - Open Aiming Points

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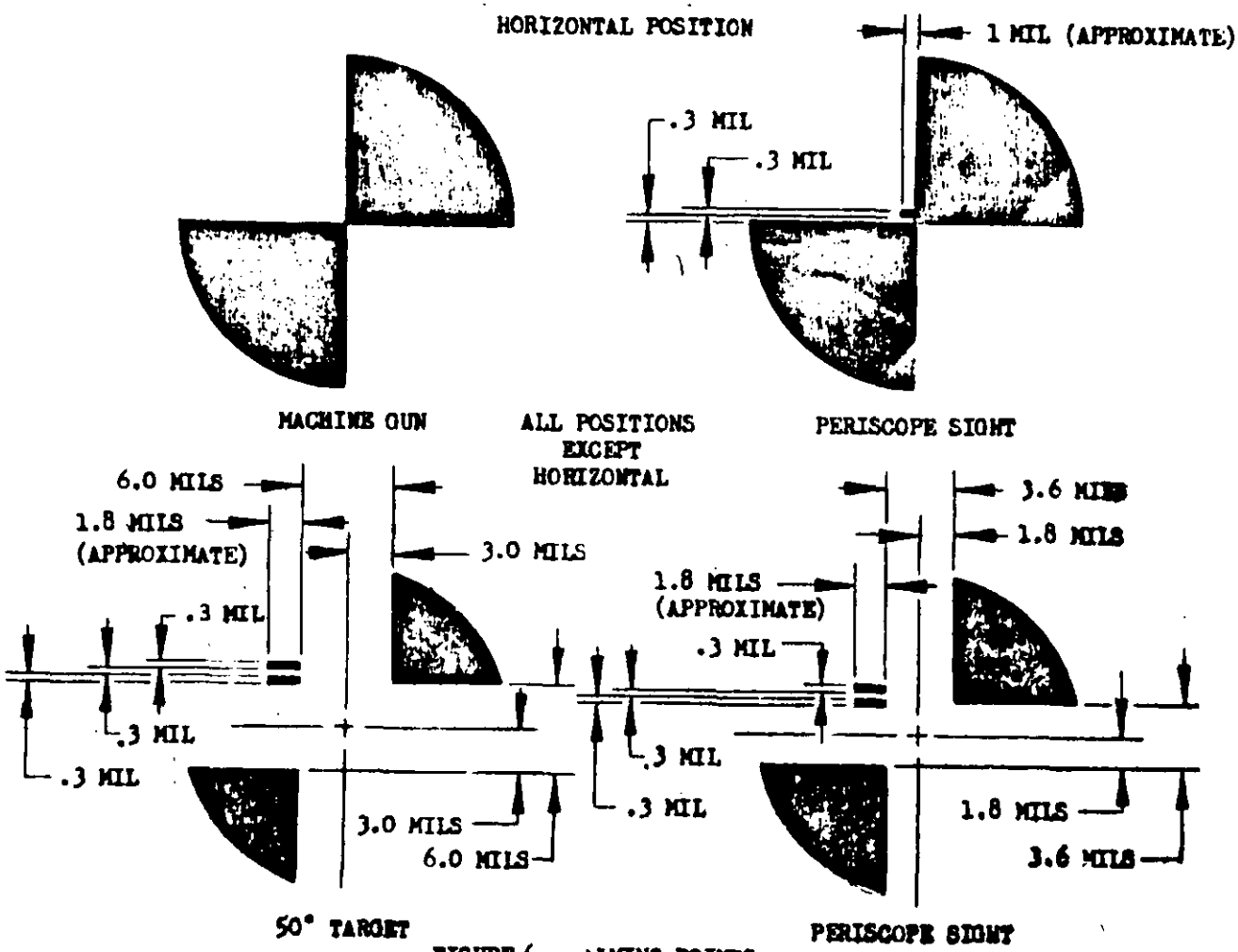


FIGURE 6 - AIMING POINTS CUPOLA MACHINE GUN AND PERISCOPE SIGHT

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DESIGN REQUIREMENTS FOR CUPOLA MACHINE GUN SIGHTING SYSTEM TARGETS

CONVERSION TABLE

NOTE: The table below is for use in converting the angular values specified on Figure 6 to dimensions on the targets.

TARGET	ANGULAR VALUE ON TARGET	REQUIRED DISTANCE IN INCHES
-10°	0.3 mil	.040
Horizontal	0.3 mil	.039
+15°	0.3 mil	.042
+30°	0.3 mil	.052
+50°	0.3 mil	.079

Horizontal aiming point for machine gun shall be located 120 ± 1/4 inches from the trunnion axis.

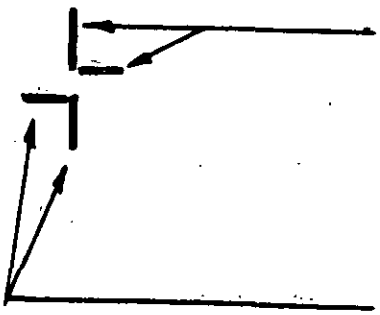
The plane of the target must be perpendicular to the horizontal line of sight through the machine gun bore within 10 minutes of arc.

Target must be adjusted so the horizontal line of sight through the machine gun bore coincides with the horizontal aiming point within .230 inches.

The centerline through the periscope targets must be parallel to the centerline through the machine gun targets within 1 minute 30 seconds of arc.

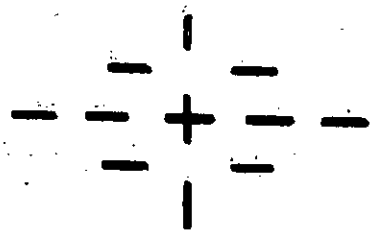
Figure 7

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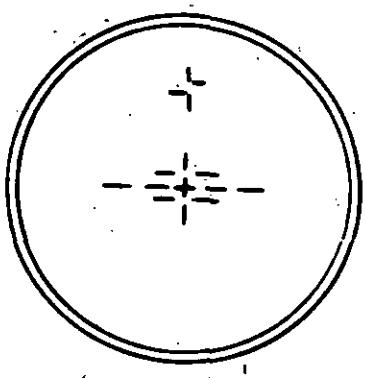
Projected thru right telescope.
Adjustments: Azimuth - ICS Knob
Elevation - Halving Knob

Projected thru left telescope.
Position relative to right telescope
using horizontal and vertical
calibrating knob.



NOTE: The cover protected halving knob shall be utilized only for the vertical adjustment of the calibrating reticle to establish a minimum recognizable separation between the upper and lower vertical reticle bars.

Illuminated Calibrating
Reticle



Range Finder M17C

Figure 8

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